IoT Document

Literature Survey, Project Identification and Selection of Devices & Components

A Real-Time Two-Wheeler Violation Detection System Using Raspberry Pi and AI-Based Computer Vision

1. IoT Lab Batch & Group Details

Field	Details
IoT Lab Batch No	A2
Academic Year	2025-2026
Programme	MCA (Master of Computer Applications)

2. Project Team Members

Sr.	Roll No.	Name
1	25MCA035	Sneh Patel
2	25MCA037	Milan Poria

3. Project Literature Survey

The following six research papers were selected from IEEE Xplore and related academic digital libraries. Each paper is directly relevant to the proposed project on real-time two-wheeler violation detection using Raspberry Pi and AI-based computer vision.

Paper 1: Two-Wheeler Vehicle Traffic Violations Detection and Automated Ticketing for Indian Road Scenario

Authors	R. S. Charran and R. K. Dubey
Published In	IEEE Transactions on Intelligent Transportation Systems, vol. 23, no. 11, pp. 22002–22007, 2022.
DOI	10.1109/TITS.2022.3186679

Summary:

This paper proposes an automated system for detecting multiple two-wheeler traffic violations on Indian roads including helmetless riding, triple riding, and illegal parking. The authors employed YOLOv5-based object detection combined with license plate recognition to identify violators and automatically generate e-challans. The system was trained and tested on a custom Indian road dataset capturing diverse lighting and road conditions. The method achieved a mean Average Precision (mAP) of over 85% for helmet detection and 80% for triple riding detection. A key limitation noted is that detection accuracy drops significantly under poor illumination and in heavy traffic where occlusion is common.

Paper 2: IoT-Enabled Intelligent System for Helmet Detection and Automated Penalty Enforcement using AI & ML

Authors	D. Mullagura et al.
Published In	Proc. 2024 International Conference on Computing, Sciences and Communications (ICCSC), IEEE, 2024.
DOI	10.1109/10988120

Summary:

This paper presents an IoT-based system that integrates a Raspberry Pi with a camera module to detect helmetless motorcycle riders at traffic signals. The system uses YOLOv5 for real-time object detection and OCR for license plate extraction. Violation data is transmitted to a centralized server via IoT protocols (MQTT), and automated penalties are enforced through SMS/email alerts. The proposed system reduces manual intervention and ensures round-the-clock monitoring. Limitations include reliance on clear number plate visibility and reduced performance in adverse weather conditions.

Paper 3: Artificial Intelligence (AI) based Detection of Traffic Violations by Two-Wheeler Vehicles

Authors	S. Anandakumar et al.
Published In	Proc. 2023 IEEE International Conference, IEEE Xplore, 2023.
DOI	10.1109/10142605

Summary:

This research presents a novel AI-based technique to electronically detect multiple two-wheeler violations on Indian roads, including helmet non-compliance, smartphone usage while riding, triple riding, wheelies, and illegal parking. The authors used a combination of deep learning models trained on a custom dataset collected from Indian urban roads. The approach also models the automated issuance of traffic tickets by correlating violations with vehicle number plates captured from surveillance footage. The system demonstrated high detection accuracy but faced limitations in scenarios with partial vehicle occlusion and in distinguishing between similar violations under low resolution.

Paper 4: Real-Time Monitoring Dashboard for Traffic Violation Detection Using Edge Computing

Authors	P. Ramakanthkumar et al.
Published In	International Journal of Advanced Information Technology (IJAIT), vol. 13, no. 2, 2023.
DOI	10.5121/ijitcs.2023.13201

Summary:

This study applies various deep learning models (YOLOv5, SSD, and Faster R-CNN) for automatic traffic violation detection using edge computing devices. The authors compare model performance in real-time scenarios and propose a lightweight model suitable for deployment on resource-constrained edge devices. The system streams violation events to a cloud-hosted monitoring dashboard accessible to traffic authorities. Experimental results showed YOLOv5 provided the best trade-off between speed and accuracy for edge deployment. Limitations include the computational constraints of edge hardware and the need for frequent model retraining as new violation categories are added.

Paper 5: AI-Based Helmet Violation Detection for Traffic Management System

Authors	Y. Said, Y. Alassaf, R. Ghodhbani et al.
Published In	Computer Modeling in Engineering & Sciences (CMES), vol. 141, no. 1, pp. 733–749, 2024.
DOI	10.32604/cmes.2024.052369

Summary:

This paper presents a deep learning-based approach to helmet violation detection using an adapted PerspectiveNet architecture, transitioning from Res2Net to the more efficient EfficientNet v2 backbone. The system was trained and validated on the India Driving Dataset (IDD), achieving an impressive detection accuracy in diverse real-world conditions. The work focuses on improving detection robustness under perspective distortion—a common challenge in road surveillance cameras. While the system demonstrated state-of-the-art performance, the primary limitation is its high computational demand during training and its dependency on high-quality annotated Indian road datasets.

Paper 6: Enhancing Traffic Violation Enforcement: A System Utilizing Helmet and Number Plate Detection

Authors	K. Patel et al.
Published In	Proc. 2024 IEEE Conference on Intelligent Transportation Systems, IEEE Xplore, 2024.

DOI	10.1109/10431231

Summary:

This paper introduces a Fine Collection System for traffic violations that combines YOLOv5-based object detection with OCR to identify helmetless riders and extract their license plate numbers. Upon detecting a violation, the system sends instant notifications via SMS or email containing details of the violation and the applicable fine. The system additionally validates the rider's license and insurance status, adjusting penalties accordingly. Repeated violations within a given timeframe trigger escalated enforcement actions. A key limitation is dependency on clear license plate visibility and the system's inability to handle multiple simultaneous violations from overlapping camera frames.

4. Problem Identification and Final Scope

4.1 Problem Statement

Helmet-less riding and triple riding are among the most prevalent traffic violations on university campuses and public roads across India. These violations significantly increase the risk of fatal road accidents. Manual monitoring by security personnel is labour-intensive, error-prone, and cannot provide continuous 24×7 surveillance. Existing manual systems cannot simultaneously monitor multiple camera feeds or automatically generate violation records. There is an urgent need for a low-cost, scalable, automated solution that can detect violations in real time, record evidence, and notify authorities without human intervention.

4.2 Final Project Scope

- Detect helmet violations in real time using AI-based object detection on a Raspberry Pi 4 edge node.
- Detect triple riding (more than two riders on a single two-wheeler) using person-counting logic within detected bounding boxes.
- Capture violation images automatically for evidentiary purposes.
- Transmit violation data securely to a cloud server using MQTT and HTTPS protocols.
- Provide an admin monitoring dashboard with real-time alerts and historical violation logs.
- Deploy the system in a campus environment for live testing and validation.

5. Project Title

A Real-Time Two-Wheeler Violation Detection System Using Raspberry Pi and AI-Based Computer Vision

6. Selection of Devices and Components

6.1 Components Used in the Six Surveyed Papers

Sr.	Paper	Key Algorithm	Hardware / Tools
1	R. S. Charran and R. K. Dubey	Deep learning with YOLOv5	YOLOv5, CCTV Camera, Custom Indian Dataset, License Plate OCR, e-Challan Module
2	D. Mullagura et al.	IoT + Edge AI on Raspberry Pi	Raspberry Pi, Pi Camera, YOLOv5, OCR Module, MQTT Protocol, Cloud Server, SMS Gateway
3	S. Anandakumar et al.	Multi-violation detection with AI	AI Model, CCTV Surveillance, Custom Indian Dataset, License Plate Reader, Ticket Issuance Module
4	P. Ramakanthkumar et al.	Edge computing with deep learning	Edge Device (Raspberry Pi / Jetson Nano), YOLOv5, SSD, Cloud Dashboard, REST API

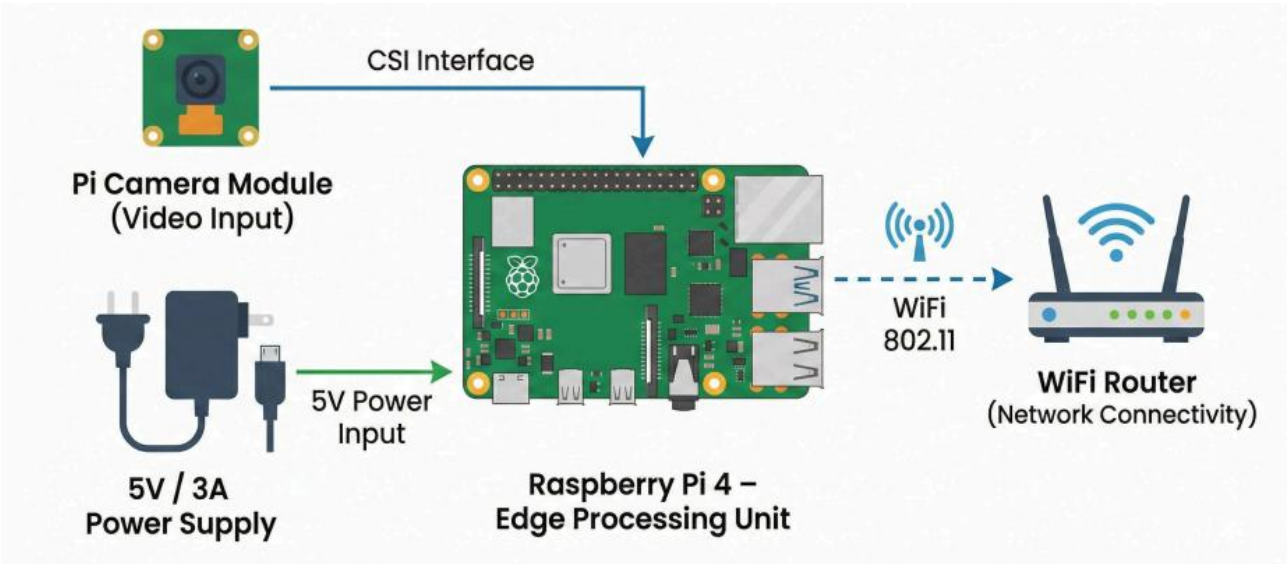
5	Y. Said, Y. Alassaf, R. Ghodhbani et al.	Custom deep learning architecture	PerspectiveNet, EfficientNet v2, India Driving Dataset (IDD), GPU-based Training
6	K. Patel et al.	Fine enforcement with YOLOv5 + OCR	YOLOv5, OCR Engine, SMS/Email Notification System, License Validation Database

6.2 Final Component List for the Proposed Project

Sr.	Component	Category	Availability	Purpose
1	Raspberry Pi 4 Model B (4 GB)	Board	IoT Lab	Edge AI processing unit
2	Pi Camera Module v2 / USB Camera	Sensor	IoT Lab	Real-time video capture
3	MicroSD Card (32 GB)	Storage	IoT Lab	OS and local violation storage
4	5V / 3A Power Supply	Power	IoT Lab	System power delivery
5	WiFi Router (802.11 b/g/n)	Network	IoT Lab	Local network connectivity
6	YOLO Object Detection Model (v5/v8)	Software / AI	Download	Helmet & person detection
7	Render	Software	Purchase / Free Tier	Data storage & dashboard
8	Admin Dashboard (Web App)	Software	Develop	Violation monitoring & alerts
9	Python 3 + OpenCV + TensorFlow Lite	Software	Open Source	Image processing pipeline

7. Concept Diagram and Working of the Proposed Solution

Circuit Diagram



The circuit diagram illustrates the complete hardware setup and data flow of the edge processing unit in the proposed system.

The Pi Camera Module serves as the primary input device and is connected to the Raspberry Pi 4 via the CSI (Camera Serial Interface). The CSI interface provides a dedicated, high-bandwidth, low-latency link that allows continuous real-time video frames to be streamed directly from the camera sensor into the Raspberry Pi's memory without burdening the USB bus. This ensures smooth and uninterrupted video capture, which is critical for real-time violation detection.

The 5V / 3A Power Supply is connected to the Raspberry Pi 4 via the USB-C power port, delivering stable and sufficient power for the board to run both the operating system and the compute-intensive AI inference model concurrently. A reliable power source is essential to prevent throttling or system instability during continuous processing.

Once the video frames arrive at the Raspberry Pi 4 Edge Processing Unit, the onboard YOLO-based AI model processes each frame locally. The model analyzes the frame to detect two-wheeler riders, identify helmet usage, and count the number of riders on a single vehicle. If a violation is detected, the system captures and stores the violation image locally on the MicroSD card along with a timestamp and violation metadata.

The processed violation data is then transmitted wirelessly from the Raspberry Pi to the WiFi Router over the WiFi 802.11 standard. The WiFi Router acts as the gateway between the edge layer and the wider network, forwarding the violation data over the internet to the cloud server via MQTT and HTTPS protocols. From the cloud server, the data is made available on the admin monitoring dashboard and alert notifications are dispatched to the relevant authorities.

In summary, the data flow is: Camera → CSI Interface → Raspberry Pi 4 (AI Processing) → WiFi 802.11 → WiFi Router → Internet → Cloud Server → Admin Dashboard & Alerts.

8. Main Functionality and Features

- Real-time helmet violation detection using AI-based object detection (YOLO model).
- Triple riding detection by counting detected persons within the two-wheeler bounding box.
- Edge AI processing on Raspberry Pi 4 to ensure low-latency, offline-capable detection.
- Automatic violation image capture with timestamped metadata for evidence.
- Cloud-based violation data storage using a PostgreSQL database for reliable record-keeping.
- Real-time monitoring dashboard for traffic authorities with live violation feeds.
- Scalable architecture deployable across multiple camera nodes on a campus network.

9. Deliverables

Sr.	Deliverable	Description
1	Functional Prototype Hardware	Raspberry Pi + Camera setup deployed at a fixed monitoring point.
2	AI Model (Trained YOLO)	Trained and quantised YOLOv8 model for helmet and person detection.
3	Edge Processing Software	Python scripts for video capture, detection, and MQTT publishing.
4	Cloud Backend & Database	FastAPI/Spring Boot server with PostgreSQL for violation storage.
5	Admin Web Dashboard	React-based dashboard showing live and historical violations.
6	Project Documentation	Complete system documentation, test reports, and user manual.

References

[1] R. S. Charran and R. K. Dubey, "Two-Wheeler Vehicle Traffic Violations Detection and Automated Ticketing for Indian Road Scenario," *IEEE Transactions on Intelligent Transportation Systems, vol. 23, no. 11, pp. 22002–22007, 2022.* DOI: 10.1109/TITS.2022.3186679

[2] D. Mullagura et al., "IoT-Enabled Intelligent System for Helmet Detection and Automated Penalty Enforcement using AI & ML," *Proc. 2024 International Conference on Computing, Sciences and Communications (ICCSC), IEEE, 2024.* DOI: 10.1109/10988120

[3] S. Anandakumar et al., "Artificial Intelligence (AI) based Detection of Traffic Violations by Two-Wheeler Vehicles," *Proc. 2023 IEEE International Conference, IEEE Xplore, 2023.* DOI: 10.1109/10142605

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[5] Y. Said, Y. Alassaf, R. Ghodhbani et al., "AI-Based Helmet Violation Detection for Traffic Management System," *Computer Modeling in Engineering & Sciences (CMES), vol. 141, no. 1, pp. 733–749, 2024.* DOI: 10.32604/cmes.2024.052369

[6] K. Patel et al., "Enhancing Traffic Violation Enforcement: A System Utilizing Helmet and Number Plate Detection," *Proc. 2024 IEEE Conference on Intelligent Transportation Systems, IEEE Xplore, 2024.* DOI: 10.1109/10431231